

LANDFORMS BY RUNNING WATER / INDIA DRAINAGE AND INTERLINKING

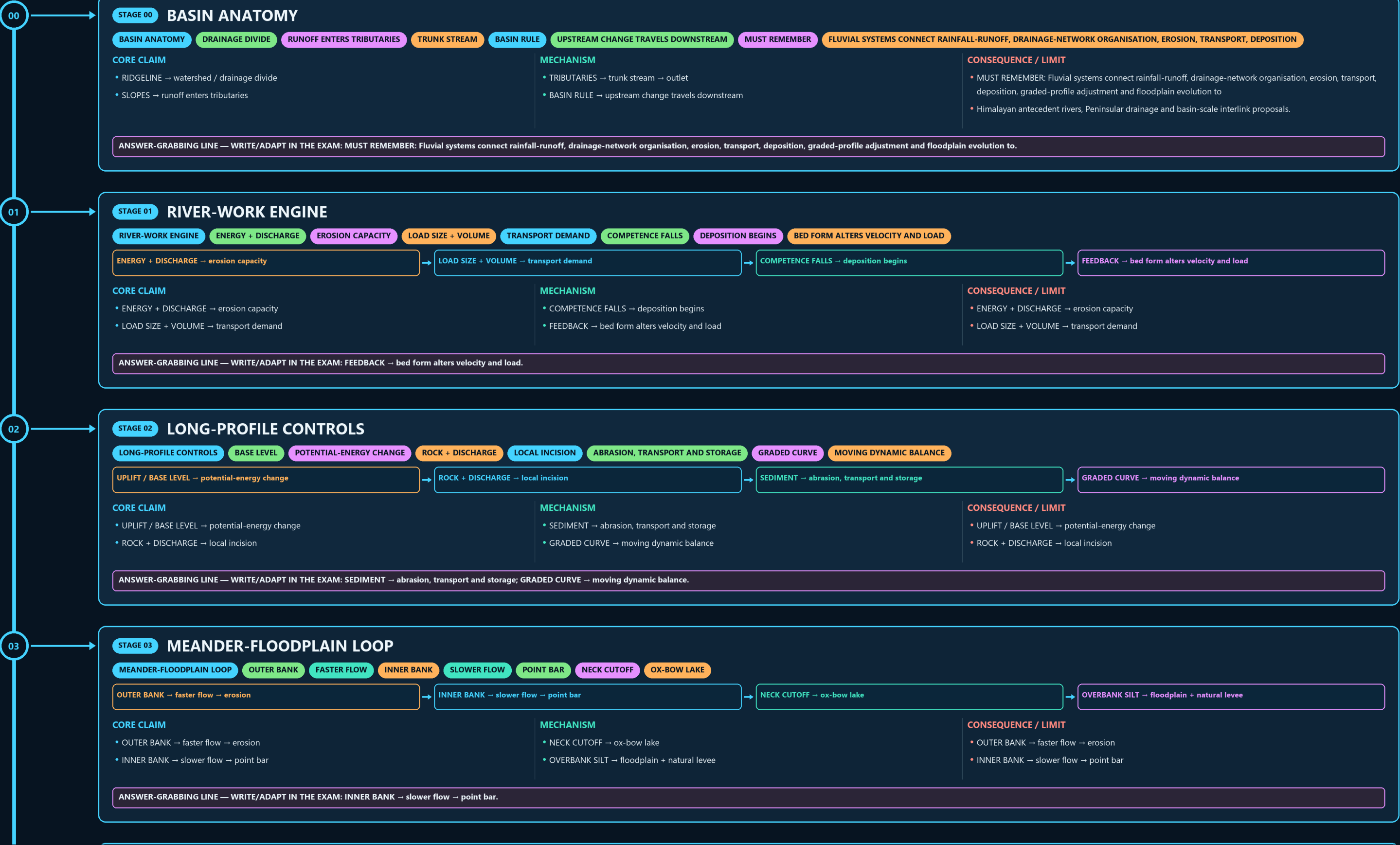
BASIN ANATOMY → RIVER-WORK ENGINE → LONG-PROFILE CONTROLS → MEANDER-FLOODPLAIN LOOP → DRAINAGE-PATTERN MATRIX → DRAINAGE-EVOLUTION FIREWALL

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • source generation g4 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles



03

STAGE 03 MEANDER-FLOODPLAIN LOOP

MEANDER-FLOODPLAIN LOOP OUTER BANK FASTER FLOW INNER BANK SLOWER FLOW POINT BAR NECK CUTOFF OX-BOW LAKE



CORE CLAIM

- OUTER BANK → faster flow → erosion
- INNER BANK → slower flow → point bar

MECHANISM

- NECK CUTOFF → ox-bow lake
- OVERBANK SILT → floodplain + natural levee

CONSEQUENCE / LIMIT

- OUTER BANK → faster flow → erosion
- INNER BANK → slower flow → point bar

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: INNER BANK → slower flow → point bar.

04

STAGE 04 DRAINAGE-PATTERN MATRIX

DRAINAGE-PATTERN MATRIX UNIFORM RESISTANCE ALTERNATING RESISTANT BANDS JOINTS AND FAULTS OUTWARD HIGH INWARD BASIN

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
DENDRITIC → uniform resistance	RECTANGULAR → joints and faults	DENDRITIC → uniform resistance
TRELLIS → alternating resistant bands	RADIAL / CENTRIPETAL → outward high / inward basin	TRELLIS → alternating resistant bands

CORE CLAIM

- DENDRITIC → uniform resistance
- TRELLIS → alternating resistant bands

MECHANISM

- RECTANGULAR → joints and faults
- RADIAL / CENTRIPETAL → outward high / inward basin

CONSEQUENCE / LIMIT

- DENDRITIC → uniform resistance
- TRELLIS → alternating resistant bands

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RADIAL / CENTRIPETAL → outward high / inward basin.

05

STAGE 05 DRAINAGE-EVOLUTION FIREWALL

DRAINAGE-EVOLUTION FIREWALL RIVER OLDER THAN UPLIFT COVER PATTERN INHERITED BELOW HEADWARD EROSION DIVERTS NEIGHBOUR FIELD CUES ELBOW-WIND GAP



CORE CLAIM

- ANTECEDENT → river older than uplift
- SUPERIMPOSED → cover pattern inherited below

MECHANISM

- CAPTURE → headward erosion diverts neighbour
- FIELD CUES → gorge / discordance / elbow-wind gap

CONSEQUENCE / LIMIT

- ANTECEDENT → river older than uplift
- SUPERIMPOSED → cover pattern inherited below

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FIELD CUES → gorge / discordance / elbow-wind gap.

06

STAGE 06 INDIA RIVER-SYSTEM CONTRAST

INDIA RIVER-SYSTEM CONTRAST HIGH RELIEF, MIXED PERENNIAL INPUTS OLD SHIELD, MONSOON-DOMINANT REGIME LONG RIVERS AND MAJOR DELTAS NARMADA-TAPI TROUGHS + SHORT GHATS STREAMS

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
HIMALAYAN → high relief, mixed perennial inputs	EAST → long rivers and major deltas	HIMALAYAN → high relief, mixed perennial inputs
PENINSULAR → old shield, monsoon-dominant regime	WEST → Narmada-Tapi troughs + short Ghats streams	PENINSULAR → old shield, monsoon-dominant regime

CORE CLAIM

- HIMALAYAN → high relief, mixed perennial inputs
- PENINSULAR → old shield, monsoon-dominant regime

MECHANISM

- EAST → long rivers and major deltas
- WEST → Narmada-Tapi troughs + short Ghats streams

CONSEQUENCE / LIMIT

- HIMALAYAN → high relief, mixed perennial inputs
- PENINSULAR → old shield, monsoon-dominant regime

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: WEST → Narmada-Tapi troughs + short Ghats streams.

07

STAGE 07 SOURCE-COURSE-MOUTH RAIL

SOURCE-COURSE-MOUTH RAIL TIBETAN HEADWATERS ARABIAN SEA DEVPRAYAG NAME BENGAL DELTA JOINED DELTA

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
HIMALAYAN → high relief, mixed perennial inputs	EAST → long rivers and major deltas	HIMALAYAN → high relief, mixed perennial inputs
PENINSULAR → old shield, monsoon-dominant regime	WEST → Narmada-Tapi troughs + short Ghats streams	PENINSULAR → old shield, monsoon-dominant regime
CORE CLAIM • HIMALAYAN → high relief, mixed perennial inputs • PENINSULAR → old shield, monsoon-dominant regime	MECHANISM • EAST → long rivers and major deltas • WEST → Narmada-Tapi troughs + short Ghats streams	CONSEQUENCE / LIMIT • HIMALAYAN → high relief, mixed perennial inputs • PENINSULAR → old shield, monsoon-dominant regime
ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: WEST → Narmada-Tapi troughs + short Ghats streams.		

07

STAGE 07 SOURCE-COURSE-MOUTH RAIL		
SOURCE-COURSE-MOUTH RAIL	TIBETAN HEADWATERS	ARABIAN SEA
DEVPRAYAG NAME	BENGAL DELTA	JOINED DELTA
INDUS → Tibetan headwaters → Arabian Sea		
GANGA → Devprayag name → Bengal delta	BRAHMAPUTRA → Tsangpo-Siang → joined delta	PENINSULA → Godavari-Krishna-Kaveri / Narmada-Tapi
CORE CLAIM • INDUS → Tibetan headwaters → Arabian Sea • GANGA → Devprayag name → Bengal delta	MECHANISM • BRAHMAPUTRA → Tsangpo-Siang → joined delta • PENINSULA → Godavari-Krishna-Kaveri / Narmada-Tapi	CONSEQUENCE / LIMIT • INDUS → Tibetan headwaters → Arabian Sea • GANGA → Devprayag name → Bengal delta
ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: BRAHMAPUTRA → Tsangpo-Siang → joined delta; PENINSULA → Godavari-Krishna-Kaveri / Narmada-Tapi.		

08

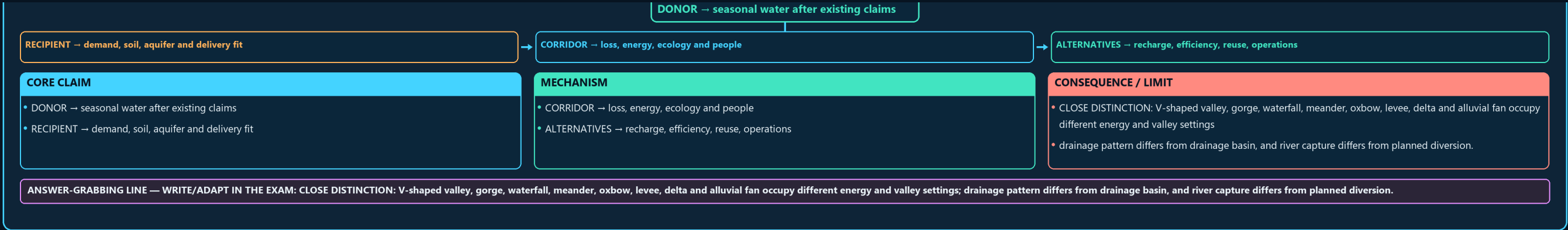
STAGE 08 DELTA BUDGET		
DELTA BUDGET	BASIN SEDIMENT DELIVERY	WAVES, TIDES AND CYCLONE SCOUR
COMPACTION + RELATIVE SEA LEVEL	DISTRIBUTARY MOBILITY OR CONFINEMENT	
CORE CLAIM - INPUT → basin sediment delivery - REMOVAL → waves, tides and cyclone scour	MECHANISM - VERTICAL → compaction + relative sea level - CHANNELS → distributary mobility or confinement	CONSEQUENCE / LIMIT - INPUT → basin sediment delivery - REMOVAL → waves, tides and cyclone scour
CORE CLAIM • INPUT → basin sediment delivery • REMOVAL → waves, tides and cyclone scour	MECHANISM • VERTICAL → compaction + relative sea level • CHANNELS → distributary mobility or confinement	CONSEQUENCE / LIMIT • INPUT → basin sediment delivery • REMOVAL → waves, tides and cyclone scour
ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VERTICAL → compaction + relative sea level; CHANNELS → distributary mobility or confinement.		

09

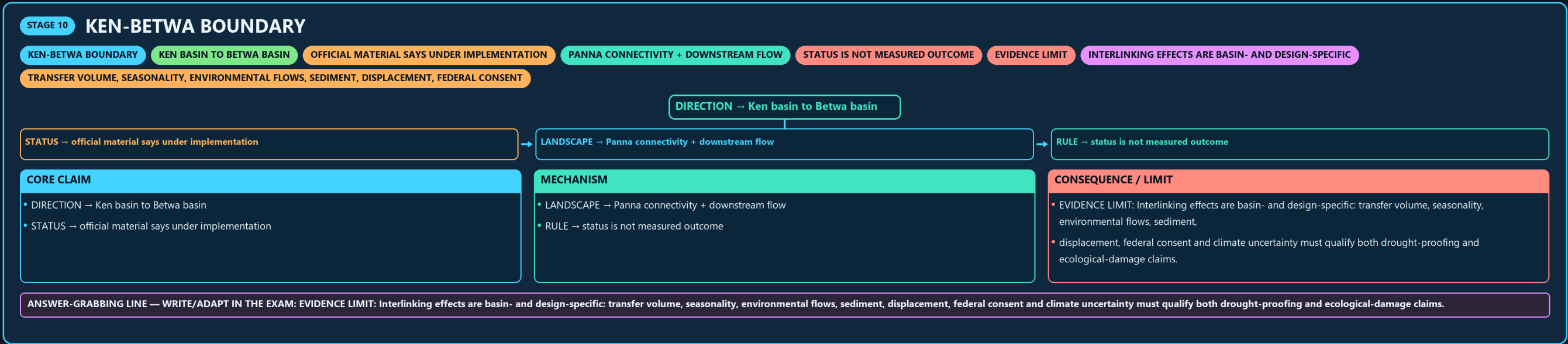
STAGE 09 TRANSFER VIABILITY TEST		
TRANSFER VIABILITY TEST	SEASONAL WATER AFTER EXISTING CLAIMS	DEMAND, SOIL, AQUIFER AND DELIVERY FIT
LOSS, ENERGY, ECOLOGY AND PEOPLE	RECHARGE, EFFICIENCY, REUSE, OPERATIONS	CLOSE DISTINCTION
V-SHAPED VALLEY, GORGE, WATERFALL, MEANDER, OXBOW, LEVEE, DELTA AND		
DONOR → seasonal water after existing claims		
RECIPIENT → demand, soil, aquifer and delivery fit	CORRIDOR → loss, energy, ecology and people	ALTERNATIVES → recharge, efficiency, reuse, operations
CORE CLAIM • DONOR → seasonal water after existing claims • RECIPIENT → demand, soil, aquifer and delivery fit	MECHANISM • CORRIDOR → loss, energy, ecology and people • ALTERNATIVES → recharge, efficiency, reuse, operations	CONSEQUENCE / LIMIT • CLOSE DISTINCTION: V-shaped valley, gorge, waterfall, meander, oxbow, levee, delta and alluvial fan occupy different energy and valley settings • drainage pattern differs from drainage basin, and river capture differs from planned diversion.
ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: V-shaped valley, gorge, waterfall, meander, oxbow, levee, delta and alluvial fan occupy different energy and valley settings; drainage pattern differs from drainage basin, and river capture differs from planned diversion.		

10

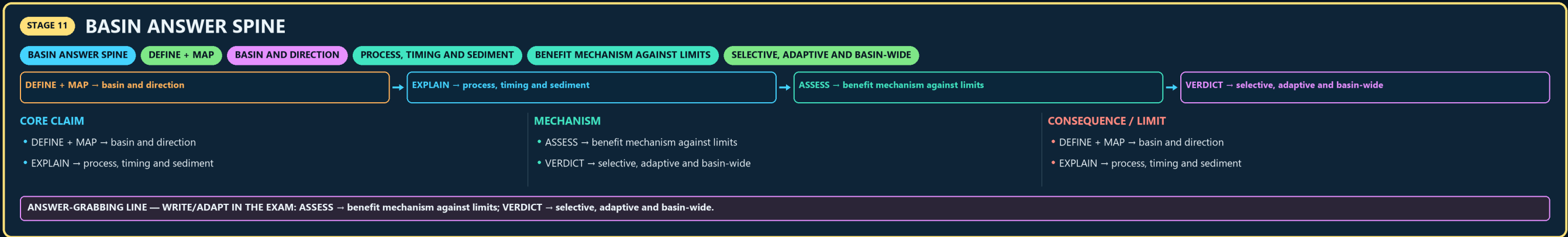
STAGE 10 KEN-BETWA BOUNDARY



10



11



E

